Sathya Kumaraguru

Portfolio | LinkedIn | Email

EDUCATION

University of California, Santa Cruz | B.S. in Computer Science |**GPA: 3.8** September 2024 - Present

California State University of Fullerton | B.S. in Computer Science | **GPA: 3.7** August 2022 – May 2024

SKILLS

- **Languages**: JavaScript (**TypeScript**), **Python**, **SQL**, C/C++, C#, Java, HTML/CSS
- **Full-Stack & Cloud**: React, Next.js, Node.js, **GCP**, Firebase, **PostgreSQL**, MongoDB, **OAuth 2.0**
- **AI & Engineering**: **TensorFlow**, Scikit-learn, **Groq Inference**, **Vercel AI SDK**, **Git**, Linux

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WORK EXPERIENCE

Overture Maps Foundation – Santa Cruz, CA | **Project Researcher** January 2025 – Present

- Collaborating with the **Overture Maps Foundation**—founded by **Amazon (AWS)**, **Meta**, and **Microsoft**—to train **Machine Learning** models that predict global business operational status with **90%+ accuracy**.
- Engineering an interactive geospatial platform using **React**, **TypeScript**, and a **Python** pipeline (**YOLO, LightGBM**) to process **Mapillary** street-level imagery for **1M+ businesses** worldwide and visualized on an **H3 hex-grid** map.

Learning Support Services – Santa Cruz, CA | **Data Structure and Algorithms Tutor** December 2025 – Present

- Provided individualized and group tutoring for **20+ undergraduate students** weekly for **Data Structures and Algorithms** course in **C** and **C++**, enhancing student comprehension of fundamental concepts like **sorting**, **searching**, and **graph theory**.

GPT Integrators – Remote | **AI Intern** August 2025 – October 2025

- Collaborated with cross-functional teams to design scalable **AI solutions** and **backend systems** that automated business processes, enhanced product integrations, and drove roughly a **20% increase** client productivity.

Lavner Education at UCLA – Los Angeles, CA | **Intern/Instructor** June 2025 – August 2025

- Delivered quality, educational instruction to elementary and middle school-aged students in a class and one-on-one environment from the provided curriculum for topics including **C++** and **Python** coding, **AI concepts**, **robotics**, and more.

Molina Healthcare – Remote | **Software Engineer Intern** June 2024 – September 2024

- Developed web-based applications with **SQL**, **C#**, and **ASP.NET** to manage 500+ healthcare claims daily, while supporting Molina Healthcare IT operations and coordinating team assignments to ensure accurate and efficient claim processing.

FEATURED PROJECTS

Cancer Mutation Analysis Platform: MutationMap

- Engineered an interactive **3D** platform using **React** and **React Three Fiber** to visualize multigene mutation patterns, integrating **Groq AI** and a **Scikit-Learn** pipeline to predict specific cancer types with **92%+ accuracy** by analyzing combined genetic data across diverse chromosomal regions.

AI Travel & Event Planning: Gopher

- Developed an AI travel assistant utilizing **Next.js** for a responsive frontend and **Groq's high-performance inference engine** to power a low-latency conversational interface, seamlessly integrating the **RESTful Google Maps** and **Ticketmaster APIs** to generate dynamic, real-time itineraries and location-based recommendations in **under 2 seconds**.

Speech-Based Language Learning: Spango

- Developed an immersive language learning platform using **React** and **Three.js**, leveraging the **Web Speech API** to enable real-time verbal interactions with **95% accuracy** and pronunciation feedback within a **3D environment**, allowing users to progress through levels primarily through **spoken conversation**.